Chutong Yang

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Research Focus

My research focuses on learning theory and algorithms for trustworthy machine learning, with an emphasis on calibration, algorithmic fairness, differential privacy, and optimization. My recent work explores LLM calibration and multi-agent collaboration. I am also working on the design and evaluation of AI agents for automated math research in both formal and natural language settings.

Education

The University of Texas at Austin, Ph.D. in Computer Science — Aug 2023 – Present
- Recipient, **Amazon AI PhD Fellowship**

Stanford University, M.S. in Computer Science — Sep 2021 – Jun 2023
- Full financial support for two years.

University of California, San Diego, B.S. in Computer Science and Mathematics — Sep 2016 – Jun 2020

Experience

Amazon Web Services (AWS), Applied Scientist Intern — Summer 2026
- Built TCSAlgBench, a benchmark consisting of research-level theoretical computer science challenge sandboxes for evaluating LLMs and agentic systems.
- Designed and evaluated agent harnesses, including agentic planning and Monte Carlo Tree Search (MCTS), and found that agentic planning performed best among the evaluated approaches.

The University of Texas at Austin, Research Assistant — Aug 2023 – Present
- Developed optimal online recalibration algorithms, with simultaneous calibration and squared-loss improvement guarantees for predictions under adversarial data sequences.
- Developed methods for LLM reasoning through uncertainty-guided multi-agent discussion and RL-based confidence calibration.

Stanford University, Research Assistant — 2022 – 2023
- Researched algorithmic fairness with Prof. Omer Reingold, focusing on multicalibration.
- Developed optimization algorithms with Prof. Aaron Sidford using interior-point methods.

Skills

Programming: Python, C++, Java, MATLAB **ML:** PyTorch

Research Software and Projects

Natural-Language Proof Verifier Benchmark Report
- Built a benchmark for evaluating LLM proof verifiers using 62 historical papers with documented proof corrections and 24 synthetic proof cases.
- Developed synthetic-proof generation tools and evaluated LLM error detection with and without access to cited prior work.

Algo2Lean – Independent collaboration with Lunjia Hu
- Contributed to a compiler that translates Python-like algorithms into Lean 4 definitions, supporting formal proofs of correctness and running-time bounds.
- Makes formalized algorithms and their output guarantees easier to read, understand, and review.

Publications

(Authors are listed alphabetically unless otherwise indicated by *.)

13. TCSAlgBench: Benchmarking Automated Proving for Research-Level Theoretical Computer Science Preprint, 2026

Chutong Yang*, Xiyuan Zhang, Yu Huang, Boran Han, Soonho Kong, Shuai Zhang, Vihang Prakash Patil, Zhen Han, Michael Bohlke-Schneider, Bernie Wang

http://arxiv.org/abs/2609.35606

12. Gap-Free Streaming PCA Beyond Rank-One Updates: Near-Optimal Rates and Applications to Differential Privacy Preprint, 2026

Anming Gu, Syamantak Kumar, Kevin Tian, ***Chutong Yang***

https://arxiv.org/abs/2609.26508

11. Optimal Recalibration of an Online Predictor **NeurIPS** 2026 (spotlight)

Lunjia Hu, Kevin Tian, ***Chutong Yang***

https://arxiv.org/abs/2607.19689

10. Simultaneous Blackwell Approachability and Applications to Multiclass Omniprediction **COLT** 2026

Lunjia Hu, Kevin Tian, ***Chutong Yang***

https://arxiv.org/abs/2602.17577

9. Spherical Leech Quantization for Visual Tokenization and Generation **CVPR** 2026 (Highlight)

Yue Zhao*, Hanwen Jiang, Zhenlin Xu, ***Chutong Yang***, Ehsan Adeli, Philipp Krähenbühl

https://arxiv.org/abs/2512.14697

8. Proportionality from Low-Dimensional Approval Data **AAMAS** 2026

Zhiyi Huang, Gregory Kehne, ***Chutong Yang***

Extended abstract: https://doi.org/10.65109/YHKI7695

7. Private Geometric Median in Nearly-Linear Time **NeurIPS** 2025

Syamantak Kumar, Daogao Liu, Kevin Tian, ***Chutong Yang***

TPDP 2025 Workshop
https://arxiv.org/abs/2505.20189

6. Omnipredicting Single-Index Models with Multi-Index Models **STOC** 2025

Lunjia Hu, Kevin Tian, ***Chutong Yang***

https://arxiv.org/abs/2411.13083

5. Testing Calibration in Nearly-Linear Time **NeurIPS** 2024

Lunjia Hu, Arun Jambulapati, Kevin Tian, ***Chutong Yang***

https://arxiv.org/abs/2402.13187

4. Omnipredictors for Constrained Optimization **ICML** 2023

Lunjia Hu, Inbal Livni-Navon, Omer Reingold, ***Chutong Yang***

https://arxiv.org/abs/2209.07463

3. Active Learning Polynomial Threshold Functions **NeurIPS** 2022

Omri Ben-Eliezer, Max Hopkins, ***Chutong Yang***, Hantao Yu

https://arxiv.org/abs/2201.09433

2. A Fast Exact Algorithm for Deployment of Sensor Nodes for Internet of Things **ISF** 2020

Qinghua Zheng*, ***Chutong Yang***, Haijun Yang, Jianhe Zhou

Information Systems Frontiers: https://link.springer.com/article/10.1007/s10796-018-9890-3

1. Detailed placement for IR drop mitigation by power staple insertion in sub-10nm VLSI **DATE** 2019

Sun ik Heo, Andrew B Kahng, Minsoo Kim, Lutong Wang, ***Chutong Yang***

https://ieeexplore.ieee.org/abstract/document/8715096

Professional Service

Conference reviewer: NeurIPS 2024, ICLR 2025, ICML 2025, ITCS 2026, NeurIPS 2026 (**Top Reviewer**), ICLR 2027